

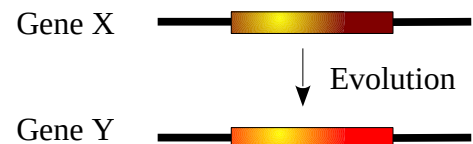
Sequences are related to each other in most case due to a common ancestor. The main forces responsible for changes in sequences are: mutations, genetic drift, and natural selection.

Homologous sequences

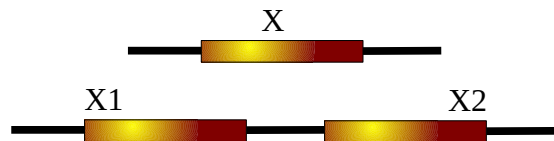
The penalties for gaps and mismatches used in algorithms for sequence alignments have to do with observed changes in DNA and protein sequences and also with the molecular structure of nucleotides and amino acids. We will start with the overall structure of sequences. Sequences with significant similarity are said to be homologous.

Homologs are defined as two or more sequences possibly from different species that are related by descent from a single common ancestral sequence. There are two main ways for sequences to have a common origin. Homologs can be either orthologous or paralogous.

One way to end up with a common ancestor is by evolution: gene X in a species is modified by a series of accumulating changes to nucleotides in the sequences to result in Gene Y and they may have similar functions (at the protein level) despite their different sequences. Homologous genes with the same function in two different species that evolved from a common ancestral gene by speciation are called **orthologs**.



Another way for two genes to be related is through the process of gene duplication. Gene duplication results in a stretch of genomic DNA that includes a gene being duplicated in the genome of an organism.



Initially the two genes X1 and X2 may be very similar to the original gene X but with evolution, X1 and X2 may “diverge” from each other into two genes Y1 and Y2 with different functions.



Homologous genes related by duplication within a genome and which may have evolved new functions are called **paralogs**.

On the other hand, two sequences that have the same function (at the protein level) that do not have enough similarity to imply an ancestral sequence are called **analogs** and may have evolved separately. Xenologs are similar sequences that resulted from genes being transferred via viruses or other means but not by evolution.

Mutations

An alignment of two nucleotide sequences can imply an evolutionary relationship between homologs. Besides the process of inheriting genes from parents, two things are needed for evolution to occur: random variation and natural selection. At the level of gene sequences, variation occurs through **mutations** in an organisms genome.

Mutations can occur in any cell in the body of an organism. As far as mutations are concerned, there are two kinds of cells we need to consider: somatic cells and germ cells. Somatic cells are the cells that make up most of the body of an organism and mutations in somatic cells, while they could be

harmful to the cells, disappear when the organism dies. Germ cells – egg cells and sperm cells in mammals – pass on genes from one generation to the next and therefore, mutations in an egg or sperm cell are inherited by an offspring.

Mutations can occur when genomic DNA in a cell is replicated. The replication of DNA occurs during cell division – one cell divides into two “daughter” cells. The two kinds of cell division are called [mitosis](#) and [meiosis](#). During mitosis a cell divides into two identical daughter cells; in the case of humans, the 23 pairs of chromosomes are replicated in each daughter cells. Mutations that occur during mitosis can cause cancer or may damage the mutated cells. Animal cell mitosis [animation](#).

Meiosis, on the other hand, is the process of generating germ cells in animals. A cell with a full set of chromosomes from both parents divides twice to produce four daughter (germ) cells, each with half as many chromosomes as the starting cells. Meiosis [animation](#). Mutations during meiosis can result in the transmission of altered genes to a zygote.

During meiosis, paired chromosomes (one from each parent) can “crossover.” This results in the two chromosomes with genes from the two parents being mixed up so that we have two chromosomes, each with a mix of genes from each parent. This is referred to as [genetic recombination](#).

The two chromatids are aligned and sections of the chromatids are exchanged. Typically many genes are exchanged between the chromatids. Misalignment of the chromatids during recombination can result in extra copies of genes in one chromatid (gene duplication) and insufficient copies in the sister chromatid (gene deletion).

Mutations that occur during mitosis in a zygote (fertilized egg cell) can result in transmission of altered genes to the future offspring of the organism that grows from the zygote. Thus, mutations that occur during either mitosis or meiosis can result in altered gene sequences in offspring.

In a fertilized egg that results in female zygote, only 24 mitotic cycles are required to start the process of setting aside her egg cells. On the other hand, sperm cells are constantly being formed in adult males. Thus, there is an increased likelihood of mutations occurring in male sperm cells than in female egg cells. On the other hand, chromosomal aberrations are more likely to occur in egg cells.

Mutations in genomic DNA sequences can occur when the proteins responsible for [replicating DNA](#) – making a copy of the DNA – make mistakes. Some of these proteins also perform a “proofreading” function and correct mistakes in replication. When both the replication and the proofreading fail, mutations occur. The three main kinds of mutations are:

1. substitution – a base is replaced by a different base
2. insertion – a base is inserted into the sequence
3. deletion – a base is deleted from a sequence

A single base can be substituted in three ways. For example:

GCCAT

GCCAT

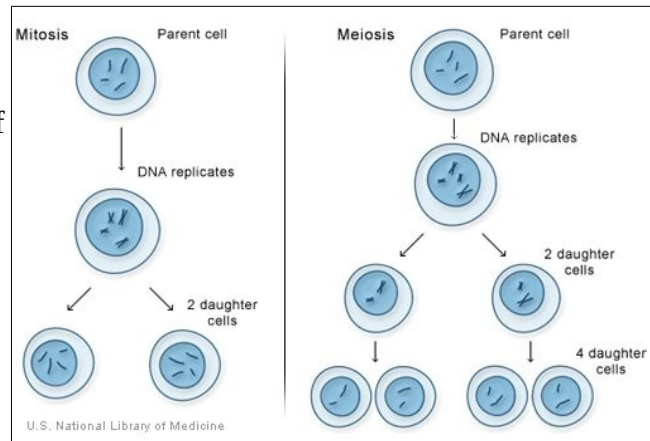
GCCAT

GCGAT

GCAAT

GCTAT

Adenines and guanines belong to a class of chemicals called “**purines**” based on their structures; while cytosines and thymines are called “**pyrimidines**”. Substituting one pyrimidine, say a C, by another (T) is not as “severe” as substituting it with a purine (an A or G) and therefore these are considered **two kinds** of substitutions: a **transition** substitution is the replacement of a purine by



another purine or a pyrimidine by another pyrimidine; a **transversion** substitution is the replacement of a purine by a pyrimidine or a pyrimidine by a purine. Single base substitutions are also called “point mutations”.

A single base insertion can occur in four ways. For example:

GCCAT	GCCAT	GCCAT	GCCAT
GCC G AT	GCC A AT	GC C TAT	GCC C AT

A single base deletion results in a single nucleotide being removed from the sequence. For example:

GCCAT
GCAT

Insertions and deletions can occur anywhere between a single base pair to thousands of base pairs. Mutations that occur in parts of the genome that do not constitute genes have no effect on the health of the organism. Those mutations that occur in the exons of genes are the ones that could potentially cause diseases.

Substitutions, insertions, and deletions can all cause inherited diseases. For example, the Cystic Fibrosis Transmembrane conductance Regulator (CFTR) gene is found on chromosome 7 and encompasses a 200,000 base pair stretch of DNA. The 27 exons of the gene contain 6132 base pairs ([NM_000492](#)) that result in a 1480 amino acid [protein](#) that is responsible for transporting chloride ions across the membranes of cells. There are descriptions ([OMIM](#)) of 136 different mutations in the CFTR gene that can cause cystic fibrosis, congenital bilateral aplasia of the vas deferens and related diseases. 90% of the cases in the US (and 66% worldwide) are caused by the deletion of 3 nucleotides that result in the loss of the amino acid phenylalanine (abbreviated “F”) at position 508 in the protein sequence. Other causes of CF are substitutions:

A point mutation of the C at position 4041 in the nucleotide sequence to a G results in a change of codon 1303 from AAC to AAG resulting in a change of the amino acid Asparagine (N) at position 1303 to a Lysine (K). This is referred to as the **N1303K** CF mutation:

	<u>“Healthy” sequence:</u>	<u>CF Mutation:</u>
	4041	4041
Nucleotide:	... AAA AAC TTG ...	... AAA A G TTG ...
Protein:	... Lys Asn Leu ...	... Lys Lys Leu ...
	1303	1303

Another point mutation at nucleotide 3978 from a G to an A results in a Tryptophan (W) becoming a stop codon (X) at amino acid position 1282. This is called the **W1282X** mutation and is the most common mutation that causes CF in the Ashkenazi Jewish population:

	<u>“Healthy” sequence:</u>	<u>CF Mutation:</u>
	3978	3978
Nucleotide:	... CAG TGG AGG ...	... CAG T G A AGG ...
Protein:	... Gln Trp Arg ...	... Gln STOP ...

This results in the truncation of the protein. The truncated protein is not functional.

Mutations are also classified on the basis of their effect on the codons or protein sequence. A “**nonsense**” mutation is mutation that results in a codon changing from one for an amino acid to one of the 3 stop codons: TAA, TAG, and TGA. The W1282X mutation above is an example of a nonsense

mutation. Translation of the mRNA that results from such a mutated sequence will result in a prematurely truncated protein; such proteins are typically non-functional.

A “**missense**” mutation results in a codon being changed and results in a different amino acid in the protein. The altered protein may be fully functional, partly functional or completely non-functional. In the case of the N1303K missense mutation described above, the protein was found to be nonfunctional.

Many mutations are “silent” in the sense that the mutation does not alter the resulting amino acid even though the codon may be changed. Remember that there are 6 codons for the amino acid Serine: thus there are 14 point mutations that are silent in the case of Serine.

A “**frameshift**” mutation can be caused by insertions or deletions. Insertions or deletions of multiples of 3 nucleotides (3, 6, 9, etc.) will only cause insertions/deletions of the corresponding codons and amino acids; neighboring amino acids are not affected and as a result, such mutations may be benign. A deletion of a single nucleotide, on the other hand, can be more devastating to an organism because the mutation can affect all the succeeding amino acid. For example, one mutation that can cause CF is the deletion of the C at nucleotide position 3659:

	<u>“Healthy” sequence:</u>						<u>CF Mutation:</u>							
	3659						3659							
Nucleotide:	...	CCT	ACC	AAG	TCA	ACC	...	...	CCT	ACA	AGT	CAA	CCA	...
Protein:	...	Pro	Thr	Lys	Ser	Thr	...	...	Pro	Thr	Ser	Glu	Pro	...

Although the deletion of nucleotide C at position 3659 does not actually change the amino acid corresponding to the codon for that nucleotide, succeeding amino acids are changed. An insertion of one or two nucleotides can also have the same consequence: a shift in the reading frame. For example, the insertion of two nucleotides (an A and a T) at position 2566 of the CFTR nucleotide sequence causes CF. In addition to the nonsense mutation shown above, frameshift mutations can also result in the creation of stop codons prematurely.

So how common are these mutations? They turn out to be pretty rare. In animals, a mutation is thought to occur once every 50 million nucleotides. This translates to about 100 to 150 mutations per cell. Since 97% of the nucleotides in our genome do not code for genes, very few genes are altered. Add in the likelihood of silent mutations and the likelihood of mutations becomes very small but not zero. Mutations do accumulate over tens of thousands of years to produce visible changes in the characteristics of a species. At the same time, some gene sequences are surprisingly similar across species. Although sequence similarity does not automatically imply homology or evolution, often such similarities point to a common ancestral sequence that may sometimes be only slightly altered by time.

Two proteins that make up hemoglobin molecules: human Hemoglobin alpha chain (Acc #P69905) and human Hemoglobin beta chain (Acc #P68871) are encoded in paralogous genes and have considerable similarities in their amino acid sequences, especially at their tail ends (called the “C-terminus”):

```

HBA      EFTPAVHASLDKFLASVSTVLTSKYR
          :::: :.. .: :.:.....: :: Degree of similarity
HBB      EFTPPVQAAYQKVVAGVANALAHKYH

```

The zebrafish Hemoglobin 1 protein is similar to an ortholog, the human Hemoglobin alpha chain protein:

Human	PAVHASLDKFLASVSTVLTSKYR
	: :.:.:.:.: . . . :. :.:
Zebrafish	PEVHVSVDKFFNNLALALSEKYR

The above two sets of protein sequence alignments are not surprising since the proteins have a common function. Often, researchers have protein sequences with unknown function and by comparing such sequences with well-understood proteins, they can begin to analyze the properties of the protein.

Scoring Matrices and the Needleman-Wunsch algorithm

The gap penalty system we introduced (page 16) can now be seen as a way to guide an alignment algorithm to the solutions that are most likely in terms of evolution. In a similar way, we may also be able to change our match and mismatch scores to take into account mutations that are more likely than others. For example, by using a match score of 1 and a mismatch score of 0, we are, effectively using the following scoring matrix:

	A	C	G	T
A	1	0	0	0
C	0	1	0	0
G	0	0	1	0
T	0	0	0	1

Such a matrix is a general way to specify match and mismatch scores when scoring an alignment. The left-most column can represent the nucleotides in one sequence and the top-most row can represent nucleotides in the other sequence. For each pair of nucleotides in an aligned sequences, we can use the above table to score that pair. An A in the first sequence and an A in the second, corresponds to a match and, thus, gets a score of 1; an A in the 1st seq and a C, G, or T in the second corresponds to the remaining entries in the first row of numbers: they are all 0s as they are considered mismatches.

Often, we may want to “discourage” transversions more than transitions. Thus, we may say that an A aligned with G will get a mismatch score of -1 while an A aligned with a C or a T will get a score of -3 since A and G are purines and C and T are pyrimidines. Similarly a C aligned with a T will get a score of -1 while a C aligned with an A or a G will get a -3. This can be specified as the following scoring matrix:

	A	C	G	T
A	1	-3	-1	-3
C	-3	1	-3	-1
G	-1	-3	1	-3
T	-3	-1	-3	1

Nucleotide scoring matrices only have 16 entries but scoring matrices for proteins will have 20 * 20, or 400 entries! In addition, with all the variations in properties of amino acids, there are many ways to score mismatches of amino acids.

A method that uses the dynamic programming structure and a scoring matrix and a linear gap penalty was proposed by the mathematicians, Saul Needleman and Christian Wunsch, in 1970. This scheme is referred to as the “[Needleman-Wunsch](#)” algorithm. This algorithm was developed at a time when protein sequences were being analyzed for similarities; not many nucleotide sequences were widely available.

The use of a scoring matrix was an integral part of the algorithm. Thus the “fill” stage of the algorithm will be slightly different: as we go along the cells of the matrix, we **set the value of each cell C_{ij} to be the maximum of the following three** possibilities (“gp” refers to the gap penalty):

- $C_{(i-1)j} + gp$
- $C_{i(j-1)} + gp$
- $C_{(i-1)(j-1)} + \text{corresponding entry from scoring matrix}$

If we take the same pair of sequences we had earlier and use a the second of the above two

scoring matrices, and a gap penalty (gp) of -1 we get the following partial scores:

	G	C	C	A	T
G	0	-1	-2	-3	-4
A	-1	1	0	-1	-2
A	-2	0	-1	-2	0
T	-3	-1	-2	-3	-1
T	-4	-2	-3	-4	-2

Write the sequence alignment corresponding to the above path.

With a gap penalty (-1) that is less severe than the mismatch score (-3), we see from the above table that we end up with alignments with many gaps. To get better alignments, we have to use more severe gap penalties or less severe mismatch scores.

Global and Local Alignments

The alignments we have been performing so far with the basic dynamic programming algorithm and the Needleman-Wunsch algorithm have been “**global**” alignments in the sense that we were comparing two **complete** sequences against each other. This may not be the ideal way to compare sequences if they have significantly different lengths; for example: AGTCGATCCG and CGA. We can see that the ideal alignment in this case would be one in which the shorter sequence is aligned exactly with a subsequence within the longer sequence:

```
AGTCGATCCG
---+++---
   CGA
```

In this case, we have “gaps” at the ends; such gaps are called “end gaps” or “terminal gaps”. The algorithms we have used so far would penalize gaps at the beginning and the end. If we were searching an entire chromosome, which may be 100 million sequences long, for a sequence of a 100 nucleotides, the terminal gaps would be enormous and it would not make any sense to perform global alignments in such a case.

One remedy would be to treat the terminal gaps differently from gaps within either sequence. For example we could set the terminal gap penalty to 0 and the nonterminal gaps to -5. Only small changes would be required in the Needleman-Wunsch algorithm to allow different gap penalties at the ends:

1. By setting the gap penalties in row 0 and column 0, we remove the gap penalties at the beginnings of both sequences.
2. By changing the algorithm so that the gap penalty, gp, is 0 in the last row and last column, we remove the gap penalties at the ends of both sequences.

This modified Needleman-Wunsch algorithm performs a “**semiglobal**” alignment by not penalizing terminal gaps.

Often we may want to compare two long sequences each with subsequences that align well. For example:

```
ATTCCAACCCAGGGACCTGCAGGCTCCCCTGCTCTCCAGATCAGACCCCTACTCTCAGAGCCCGGT
and
GCCAGGGATTTCCGTCCCCTGCTCTCAGAGGACCCTACTCTCAGAGAA
```

These two sequences share a number of subsequences:

```
ATTCCAACCCAGGGACCTGCAGGCTCCCCTGCTCTCCAGATCAGACCCCTACTCTCAGAGCCCGGT
+++++++
GCCAGGGATTTCC_____GTCCCCTGCTCTCAGAG_____GACCC_TACTCTCAGAGAA
```